

Robotic Deployment Responsibility Assignment Matrix

Lunar Surface Infrastructure Program

ACCOUNTABILITY EXECUTION DECISION RIGHTS

Owner	Program Director	Status	Draft for review
Version	0.1	Basis	Level 4 Work Packages v0.1
Coverage	30 work packages	Date	22 September 2026

Purpose: Establish one accountable function, the required execution functions, consultation paths, information flow, and decision authority for each Level 4 work package within WBS element 1.3 Robotic Deployment and Site Preparation.

Management conclusion: The matrix assigns operational ownership without transferring design authority from the Habitat Program, Lunar Data Center Program, or Shared Infrastructure Owners. Cross-branch acceptance remains a formal decision-gate action.

Director Decision and Intended Use

Approve this draft as the organizational basis for detailed planning of robotic deployment and site preparation. Approval authorizes the program team to nominate individuals, resource the accountable functions, build the integrated schedule, and negotiate cross-branch commitments. It does not authorize execution, procurement, launch, or crew arrival.

What this matrix controls

- Ownership of work-package scope, schedule, cost, risk, interfaces, and completion evidence.
- The functions that perform the work and the organizations that must be consulted before decisions are finalized.
- The stakeholders that receive status and decision outcomes without becoming approval authorities.
- The authority and evidence required to close each work package at a defined decision gate.

Program boundaries

Robotics owns the means and evidence for survey, preparation, installation, inspection, and remote support. Habitat, data-center, and shared-infrastructure branches retain design authority for their systems and accept the applicable site, interface, installation, commissioning, and readiness evidence. Launch transportation and crew authorization remain outside WBS element 1.3 except where explicitly identified as an interface.

Control status

This version is a proposed organizational baseline. Named individuals, labor estimates, calendar dates, supplier assignments, and contractual commitments will be added only after branch-owner review and program authorization.

RACI Operating Rules

The matrix is designed for program execution, not merely for communication. Each designation establishes a distinct management expectation.

A Accountable	One function only. Owns the outcome, integrated performance, decision preparation, and objective closure evidence.
R Responsible	Performs or directly manages the work. Multiple functions may be responsible when execution is integrated.
C Consulted	Provides required two-way technical, operational, safety, or customer input before decisions and baseline changes.
I Informed	Receives controlled status, decisions, and changes. This is a one-way information obligation.

Governance constraints

- A work package cannot be baselined without one named accountable manager and confirmed responsible functions.
- A design authority may accept evidence without becoming accountable for robotic execution.
- The decision authority may approve, conditionally approve, reject, or return a package for corrective action.
- Unresolved safety-critical or mission-critical findings prevent closure unless the authorized risk authority formally accepts them.
- Changes to A, decision authority, acceptance evidence, or work-package boundaries require configuration-controlled approval.
- Cross-branch conflicts that cannot be resolved at the branch level escalate to the Program Director with documented options and impacts.

Functional Role Dictionary

Codes in the matrix identify organizational functions, not specific people. Individual appointments should be recorded in an approved staffing register or integrated program organization plan.

Code	Function	Program role
PD	Program Director	Owns program outcomes, cross-branch priorities, baseline authorization, and escalated decisions.
PMO	Program Controls and Integration	Integrates scope, schedule, cost, risk, configuration, and performance reporting.
SE	Systems Engineering and Integration	Controls requirements, architecture, interfaces, verification logic, and technical integration.
ROB	Robotics Engineering and Autonomy	Develops and operates robotic systems, autonomy, tooling, mobility, and fleet capabilities.
SCI	Site Science and Geotechnical	Owns survey science, terrain, regolith, environment, hazard characterization, and site evidence.
NET	Communications Navigation and Cyber	Provides surface communications, positioning, timing, cybersecurity, and network operations.
SUR	Surface Engineering and Construction	Plans and executes work-zone control, clearing, grading, foundations, utilities, and assembly.
LOG	Launch Transportation and Logistics	Coordinates manifests, delivery interfaces, cargo handling, staging, spares, and replenishment.
MA	Mission Assurance Quality and Configuration	Controls quality evidence, qualification, nonconformance, configuration, and readiness assurance.
SAFE	Safety Reliability and Resilience	Assesses hazards, reliability, failure response, residual risk, and operational resilience.
MOPS	Mission Operations and Ground Segment	Controls command authority, operating procedures, monitoring, diagnostics, and remote recovery.
HAB	Habitat Program	Retains habitat design authority and accepts habitat-related sites, interfaces, installation, and readiness evidence.
DCO	Lunar Data Center Program	Retains data-center design authority and accepts related interfaces, installation, and readiness evidence.
SIO	Shared Infrastructure Owners	Retain design and acceptance authority for common power, thermal, utilities, and other shared services.

1.3.1 Robotic Fleet and Payload Configuration

Branch outcome: A qualified robotic fleet and payload configuration capable of executing the approved survey, preparation, installation, inspection, and contingency missions.

Branch gate: Robotic Fleet Readiness Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.1.1	Robotic Mission Concept of Operations and Requirements	Robotics Systems Engineering Lead	SE, ROB	PMO, SAFE, MOPS, HAB, DCO, SIO	PD, LOG
1.3.1.2	Fleet Architecture and Role Allocation	Robotic Fleet Architecture Lead	ROB, SE	LOG, NET, MOPS, SAFE	PD, HAB, DCO, SIO
1.3.1.3	Payload Tooling and Interface Configuration	Robotic Payload Integration Lead	ROB, LOG	SE, MA, HAB, DCO, SIO	PD, MOPS
1.3.1.4	Mobility Autonomy and Fault Management Verification	Robotics Verification Lead	ROB, NET	SE, SAFE, MA, MOPS	PD, LOG
1.3.1.5	Integrated Fleet Qualification and Launch Readiness	Robotics Mission Assurance Lead	MA, ROB, LOG	SE, SAFE, PMO	PD, HAB, DCO, SIO, MOPS

Director Review Focus

Confirm the fleet architecture closes all required surface missions, the interface owners support the payload configuration, and qualification evidence is aligned to launch-integration entry criteria.

1.3.1 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.1.1	Robotic Mission Concept Review	Program Director	Approved concept and allocated requirements baseline
1.3.1.2	Fleet Architecture Review	Robotics Branch Manager	Approved fleet composition and role allocation
1.3.1.3	Payload Interface Review	Interface Control Board	Controlled tooling and interface configuration
1.3.1.4	Mobility and Autonomy Verification Review	Chief Systems Engineer	Verified mobility autonomy and fault management evidence
1.3.1.5	Robotic Fleet Readiness Review	Program Director	Qualified fleet and launch-readiness recommendation

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

1.3.2 Lunar Site Survey and Hazard Characterization

Branch outcome: A controlled evidence package characterizing terrain, environment, hazards, and site suitability for landing, traversal, construction, and sustained operations.

Branch gate: Lunar Site Characterization Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.2.1	Survey Data Requirements and Campaign Plan	Site Characterization Lead	SCI, ROB	SE, LOG, HAB, DCO, SIO	PD, NET, SAFE
1.3.2.2	Terrain Topography and Geotechnical Characterization	Lunar Geotechnical Lead	SCI, ROB	SUR, SAFE, MA, SE	PD, HAB, DCO, SIO
1.3.2.3	Illumination Thermal Radiation and Dust Characterization	Lunar Environment Lead	SCI	SAFE, SE, HAB, DCO, SIO	PD, ROB, MOPS
1.3.2.4	Landing Traverse and Work Zone Hazard Mapping	Surface Hazard Analysis Lead	SCI, SUR	LOG, ROB, SAFE, NET, MOPS	PD, HAB, DCO, SIO
1.3.2.5	Site Suitability Recommendation and Baseline	Site Selection Integration Lead	SE, SCI	PD, LOG, SAFE, HAB, DCO, SIO	ROB, NET, SUR, MOPS

Director Review Focus

Confirm every site decision criterion has objective evidence, uncertainty is visible, and infrastructure owners accept the environmental and geotechnical inputs used for design.

1.3.2 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.2.1	Survey Campaign Readiness Review	Chief Systems Engineer	Approved survey requirements and campaign plan
1.3.2.2	Terrain and Geotechnical Data Review	Site Characterization Lead	Validated terrain and geotechnical dataset
1.3.2.3	Lunar Environment Characterization Review	Chief Systems Engineer	Validated illumination thermal radiation and dust profiles
1.3.2.4	Surface Hazard Map Review	Surface Operations Authority	Controlled landing traverse and work-zone hazard maps
1.3.2.5	Lunar Site Characterization Review	Program Director	Approved site-suitability recommendation and evidence baseline

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

1.3.3 Surface Communications and Navigation Setup

Branch outcome: An operational surface communications, navigation, and timing capability supporting robotic preparation and later infrastructure operations.

Branch gate: Surface Network Readiness Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.3.1	Surface Communications and Navigation Architecture	Surface Network Architecture Lead	NET, SE	ROB, MOPS, SAFE, SIO	PD, HAB, DCO, LOG
1.3.3.2	Relay and Base Station Deployment	Surface Network Deployment Lead	NET, ROB, SUR	LOG, SIO, MA	PD, MOPS, HAB, DCO
1.3.3.3	Position Navigation and Timing Establishment	Surface Navigation Lead	NET, SCI	ROB, MOPS, SE	PD, SUR, LOG
1.3.3.4	Network Integration Cybersecurity and Contingency	Surface Network Operations Lead	NET, MOPS	ROB, SAFE, SE, MA	PD, HAB, DCO, SIO
1.3.3.5	Coverage Performance Verification and Handover	Surface Network Verification Lead	NET, MA	MOPS, ROB, SE	PD, HAB, DCO, SIO, SUR

Director Review Focus

Confirm communications, navigation, timing, cybersecurity, degraded modes, and operational handover support autonomous work before surface construction depends on the network.

1.3.3 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.3.1	Surface Network Architecture Review	Chief Systems Engineer	Approved communications navigation and timing architecture
1.3.3.2	Network Deployment Readiness Review	Surface Network Manager	Installed and powered relay and base-station assets
1.3.3.3	Position Navigation and Timing Acceptance Review	Surface Network Manager	Operational local position navigation and timing reference
1.3.3.4	Network Integration and Contingency Review	Cybersecurity and Operations Authority	Secure integrated network and contingency capability
1.3.3.5	Surface Network Readiness Review	Program Director	Verified coverage performance and operations handover

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

1.3.4 Site Preparation and Grading

Branch outcome: Prepared, stabilized, and verified surface areas and routes suitable for infrastructure installation and robotic operations.

Branch gate: Prepared Site Acceptance Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.4.1	Site Preparation Plan and Equipment Readiness	Site Preparation Manager	SUR, ROB	PMO, SAFE, LOG, SIO	PD, SCI, NET, HAB, DCO
1.3.4.2	Work Zone Demarcation and Hazard Controls	Surface Safety and Work Control Lead	SUR, SAFE	SCI, ROB, NET, MOPS	PD, LOG, HAB, DCO, SIO
1.3.4.3	Obstacle Removal and Surface Clearing	Robotic Construction Lead	SUR, ROB	SCI, SAFE, MA, MOPS	PD, LOG, HAB, DCO, SIO
1.3.4.4	Grading Compaction and Foundation Preparation	Lunar Civil Works Lead	SUR, ROB	SCI, MA, SE, HAB, DCO, SIO	PD, LOG, SAFE, MOPS
1.3.4.5	Prepared Site Acceptance Survey and Handover	Prepared Site Acceptance Lead	MA, SUR, SCI	SE, HAB, DCO, SIO	PD, ROB, LOG, SAFE, MOPS

Director Review Focus

Confirm site preparation is sequenced against equipment availability, hazard controls, and acceptance criteria supplied by the receiving infrastructure owners.

1.3.4 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.4.1	Site Preparation Readiness Review	Surface Construction Authority	Approved execution plan and ready equipment
1.3.4.2	Work Zone Control Review	Surface Operations Authority	Established work zones routes and hazard controls
1.3.4.3	Surface Clearing Acceptance Review	Site Preparation Manager	Cleared work areas and travel corridors
1.3.4.4	Civil Works Verification Review	Chief Systems Engineer	Verified graded compacted and dimensionally controlled features
1.3.4.5	Prepared Site Acceptance Review	Program Director	Accepted as-built site baseline and handover package

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

1.3.5 Infrastructure Deployment and Assembly

Branch outcome: Robotic deployment, placement, assembly, and installation evidence for program infrastructure at accepted lunar surface locations.

Branch gate: Infrastructure Installation Readiness Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.5.1	Deployment Sequence and Robotic Assembly Planning	Surface Installation Manager	SUR, ROB, PMO	LOG, SE, HAB, DCO, SIO	PD, NET, SAFE, MOPS, MA
1.3.5.2	Cargo Offload Transport and Staging	Lunar Logistics Lead	LOG, ROB	MA, SUR, HAB, DCO, SIO	PD, SE, SAFE, MOPS
1.3.5.3	Utility Pathway and Interface Installation	Surface Utilities Installation Lead	SUR, ROB	SIO, NET, MA, SE	PD, HAB, DCO, LOG, MOPS
1.3.5.4	Equipment Placement Anchoring and Assembly	Robotic Assembly Lead	SUR, ROB	MA, SE, HAB, DCO, SIO	PD, LOG, SAFE, MOPS
1.3.5.5	Installation Verification and System Handover	Installation Verification Lead	MA, SE	SUR, HAB, DCO, SIO	PD, ROB, LOG, SAFE, MOPS

Director Review Focus

Confirm cargo arrival, staging, utilities, robotic handling, physical installation, and receiving-owner acceptance form one integrated and recoverable installation sequence.

1.3.5 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.5.1	Deployment and Assembly Sequence Review	Program Director	Approved integrated installation sequence
1.3.5.2	Cargo Receipt and Staging Review	Surface Installation Manager	Accounted inspected and staged cargo baseline
1.3.5.3	Utility Interface Installation Review	Shared Infrastructure Design Authority	Verified utility pathways tie points and as-built records
1.3.5.4	Equipment Installation Verification Review	Receiving Design Authorities	Verified placement anchoring assembly and protection
1.3.5.5	Infrastructure Installation Readiness Review	Receiving Branch Owners	Accepted installation baseline and commissioning handover

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

1.3.6 Inspection Maintenance and Remote Readiness Verification

Branch outcome: A remotely supportable surface system with established condition baselines, maintenance capability, anomaly closure, and evidence supporting crew-arrival decisions.

Branch gate: Uncrewed Surface Readiness Review

Responsibility Assignment

WBS	Work package	Accountable	Responsible	Consulted	Informed
1.3.6.1	Inspection and Condition Baseline	Remote Inspection Lead	MA, ROB	SUR, SIO, HAB, DCO, MOPS	PD, SE, SAFE, LOG
1.3.6.2	Preventive Maintenance Spares and Repair Readiness	Surface Maintenance Lead	ROB, LOG	SAFE, HAB, DCO, SIO, PMO	PD, MA, SE, MOPS
1.3.6.3	Remote Command Diagnostics and Recovery Validation	Remote Operations Readiness Lead	MOPS, NET, ROB	SAFE, SE, HAB, DCO, SIO	PD, MA, LOG
1.3.6.4	Anomaly Resolution and Configuration Closure	Mission Assurance and Configuration Lead	MA, SE	PMO, SAFE, HAB, DCO, SIO	PD, ROB, NET, SUR, LOG, MOPS
1.3.6.5	Uncrewed Readiness Demonstration and Crew Arrival Recommendation	Integrated Readiness Lead	MOPS, SE, MA	PD, SAFE, HAB, DCO, SIO, ROB	LOG, NET, SUR, SCI, PMO

Director Review Focus

Confirm the program can observe, diagnose, maintain, recover, and demonstrate the integrated surface system before recommending progression toward crew arrival.

1.3.6 Decision Gates and Acceptance

A work package closes only when the accountable function presents the required evidence and the designated decision authority records disposition. Conditional approval must identify the residual action, owner, due date, and operational constraint.

WBS	Decision gate	Decision authority	Acceptance evidence
1.3.6.1	Initial Condition Baseline Review	Mission Assurance Authority	Controlled initial condition baseline
1.3.6.2	Remote Maintenance Readiness Review	Surface Operations Authority	Approved maintenance spares tooling and repair capability
1.3.6.3	Remote Diagnostics and Recovery Review	Mission Operations Authority	Validated command diagnostics isolation and recovery capability
1.3.6.4	Anomaly and Configuration Closure Review	Program Director	Closed or formally accepted anomalies and controlled configuration
1.3.6.5	Uncrewed Surface Readiness Review	Program Decision Authority	Accepted readiness recommendation for the next program phase

Escalation path

The accountable function first resolves issues within the branch. Cross-branch interface, resource, safety, configuration, or schedule conflicts move to Systems Engineering and Program Controls for integration. The Program Director decides unresolved tradeoffs that affect program baselines, mission objectives, or readiness commitments.

Baseline Approval and Next Actions

The Program Director should authorize this matrix only after each branch confirms that accountability is singular, responsible organizations are resourced, consultation obligations are practical, and decision authorities are empowered to accept the specified evidence.

Required review actions

- Confirm the functional assignments for all 30 work packages and resolve any duplicate or missing ownership.
- Nominate individuals for each accountable function and identify alternates for mission-critical packages.
- Validate that Habitat, Lunar Data Center, and Shared Infrastructure design authorities accept the stated boundaries and evidence responsibilities.
- Convert the RACI assignments into staffing demand, control-account ownership, and integrated schedule responsibility.
- Link each decision gate to predecessor milestones, required entrance criteria, planned review dates, and closure actions.
- Baseline the matrix through program configuration control and review it after architecture, site, supplier, or mission-sequence changes.

Recommended subsequent artifacts

Develop the preliminary integrated milestone network, milestone dictionary, interface responsibility register, and staffing plan. Together, these artifacts will convert the approved work-package structure and responsibility model into an executable program-control framework.

Revision Record

Version	Date	Status	Description
0.1	22 Sep 2026	Draft for review	Initial RACI and decision-rights assignment for 30 Level 4 robotic-deployment work packages.